

Momentum factor investing: Evidence and evolution

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Abstract

Momentum is a foundational factor in equity markets. We review its evolution in the literature and analyze the momentum factor empirically across a wider variety of tests. Our empirical analyses demonstrate robust empirical support for the momentum factor over domestic and global stock markets spanning up to 150 years of data and a wide variety of design choices, establishing momentum's resilience against data mining and arbitrage concerns. Momentum has transitioned from pure price-based trends to advanced fundamental, firm-specific, and network-based trends that improve the effectiveness of the momentum factor. Finally, momentum is exposed to crash risk, but we find that risk-managed momentum strategies mitigate the crash risk and improve the risk efficiency of the momentum factor. Overall, the momentum factor premium is sizable, robust, persistent, and fundamentally multi-dimensional.

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1 Introduction

Momentum is one of the most widely studied equity factors, as well as most prominent factors exploited by investors. Momentum is a term originating from physics referring to the tendency of an object moving in one direction through force of motion, to continue moving in that direction. In investing, we refer to momentum as the tendency of assets that have exhibited strong historical trends to continue those trends and hence outperform in the future — commonly summarized as “winners keep winning, losers keep losing.”

Akin to the field of physics, from the early introduction of momentum till now, academic and industry professionals have been working not only to identify momentum patterns, but also to increase the fundamental understanding of it. With the body of momentum literature starting in 1967, the first concept of momentum showed the effectiveness of relative strength indicators in distinguishing future winners from losers (Levy, 1967). Next, De Bondt and Thaler (1985) documented both long-term reversal patterns and the persistence in 12-month winners and the seminal study by Jegadeesh and Titman (1993) significantly elevated momentum’s prominence, leading to an exponential increase in academic investigations into the momentum phenomenon. Moreover, momentum has been documented beyond equities, appearing consistently across diverse asset classes such as corporate bonds, equity futures, bond markets, currencies, and commodities (Baltussen, Swinkels, & Van Vliet, 2021).

Momentum strategies have also become prominent among investment practitioners, with substantial assets under management employing these strategies, including multi-factor and more general systematic alpha approaches.

In this paper, we review the historical evolution of momentum — from initial studies on relative strength indicators to recent studies on network-driven momentum — and examine

its evidence empirically. We show that momentum is a persuasive phenomenon present across the bulk of major stock markets and spanning more than 150 years of investment data, and show how it moves beyond pure price-based measures of momentum into fundamental, peer-based, and risk-managed approaches.

We start with reviewing the initial empirical evidence on traditional price momentum, foundational papers, and key empirical findings. Subsequently, we examine the robustness of price momentum effects, to rule out explanations related to data mining or arbitrage and address out-of-sample decay, including international evidence and deep sample evidence. Next, we address the evolution of momentum beyond purely price-based measures, considering fundamental momentum such as earnings momentum and analyst revisions, firm-specific enhancements, industry momentum, network-based momentum, and factor momentum. Additionally, the paper evaluates risk considerations of momentum strategies, emphasizing crash risk and exploring crash-managed momentum strategies designed to significantly enhance risk-efficiency. Finally, the paper discusses economic drivers of momentum and concludes by summarizing key insights and implications for investors and researchers.

2 Price Momentum: initial evidence

In this section, we review the empirical evidence for the momentum effect, which goes back all the way to the very first empirical asset pricing studies in the nineteen sixties, and which by now consists of an extensive stream of literature. Our primary focus is what we term price momentum, or momentum in past stock returns that continue in the future. We first discuss the initial historical studies on momentum, followed by a replication of the momentum effect

based on the sample period used in the seminal momentum paper by Jegadeesh and Titman (1993).

2.1 A history of Momentum

The first papers on momentum focused on examining whether returns could be predicted with past returns, in contrast to the idea of efficient and rational markets as summarized in the Efficient Market Hypothesis (EMH; Fama (1960)). Past price changes are easy-to-observe public information that should not easily predict future returns. Early research on momentum includes Levy (1967), who analyzed weekly prices of 200 NYSE-listed stocks, ranking them by relative strength defined as the stock's price change relative to a benchmark index. He found that stocks that were in the top relative strength tier over a 26-week period tended to continue outperforming over the subsequent 26 weeks, demonstrating the effectiveness of relative strength indicators in distinguishing future winners from losers.

Subsequent work by De Bondt and Thaler (1985) empirically test whether behavioral biases, specifically overreaction, influence security prices. Their primary focus is documenting a long-term reversal — loser stocks over the past three to five years outperform winner stocks in the following three years. Although not their primary focus, they also documented persistence in past 12-month winners versus losers - or momentum. Their table I and figure 2 show that past year winners outperform past year loser stocks over the next year by 7.6%, on average.

The seminal study by Jegadeesh and Titman (1993) (henceforth JT) significantly elevated momentum's prominence, leading to a substantial increase in academic investigations into this phenomenon. With the increasing availability of data and computing power, JT analyzed

all NYSE and AMEX common stocks in the CRSP database from 1965 to 1989. Each month, they ranked stocks by past 3-, 6-, 9-, and 12-month cumulative returns, and created zero-cost momentum portfolios that went long the winner stocks and short the losers. These portfolios were then held for matching horizons of 3–12 months. Their results showed that momentum portfolios produced statistically significant abnormal returns that could not be explained by risk-adjusted models of the time, and were robust across sub-periods and specifications. Figure 1 summarizes the ‘evolution of momentum’ since these early studies, focusing on the key developments we will discuss in this study.¹

[Figure 1 about here]

2.2 Empirical evidence

We start our empirical analyses by examining momentum used over the sample period used in the seminal paper by JT, but defining past winner or loser stocks based on their price momentum over the previous 12 months excluding the current month and holding these portfolios for one month - we would argue the standard definition across most academic studies. To this end, we use the monthly U.S. stock returns data from Jensen, Kelly, and Pedersen (2023), who obtained this data from the Center for Research in Security Prices (CRSP) for stocks listed on the NYSE, AMEX, and NASDAQ. For each month in our sample, we create quintile breakpoints for quintile portfolio sorts using non-microcap stocks based on their price momentum over the previous 12 months, excluding the current month.²

¹We like to stress that the timeline and studies we discuss in this paper are by no means complete. We have made choices to keep the scope limited and have therefore omitted the majority of academic studies written on momentum.

²In line with Jensen et al. (2023), we define microcap stocks as stocks with a market cap below the twentieth percentile of the NYSE.

Using these breakpoints, we assign all stocks, including microcaps, to their corresponding quintiles, following the methodology of Jensen et al. (2023). Next, we compute the value-weighted (VW) and equally weighted (EW) portfolio returns of each quintile, and subtract the bottom quintile from the top quintile to obtain the winner-minus-loser portfolio ('5-1').

Table 1 contains the results. We find that the losers portfolio (portfolio '1') realized an average return of 6.88% per annum, increasing monotonically to 18.16% for winner portfolios ('5'). The difference winner-minus-loser portfolio ('5-1') realized an average return of 11.28% per annum between 1965 and 1989. This return spread is highly significant with a t-statistic of 3.50. In other words, past winner stocks outperform past losers, the momentum effect.

These results become stronger when equal-weighting stocks, with winners outperforming losers by a margin of 13.38% (t-statistic = 4.78), indicating that momentum returns are even stronger under smaller caps. This is in line with earlier results of Jegadeesh and Titman (2001).

[Table 1 about here]

The momentum premium is persistent over the JT sample period, as shown in the yellow highlighted period in figure 2. Return spreads between the winner and loser portfolio increase almost monotonically over the JT sample period except for a small dip in equally weighted (not value-weighted) momentum returns during the mid 1970s.

[Figure 2 about here]

3 Momentum is a very robust phenomenon

As mentioned above, momentum has been one of the most widely studied equity factors. How robust is the effect documented by JT and others? In this section, we examine the robustness of price momentum effects by studying its empirical evidence over the sample period before and after the initial studies, evidence across international markets, further out-of-sample evidence over deep-sample studies since the 19th century, and potential out-of-sample decay of momentum’s profitability.

3.1 An update of Momentum’s evidence

Since the seminal JT study, several studies empirically examine the robustness of momentum to other factors and over additional sample periods (e.g., Jegadeesh and Titman (2001); Asness, Moskowitz, and Pedersen (2013)). Fama and French (1996) shows the momentum factor cannot be explained by the Fama-French three factor model — consisting of the market, size, and book-to-market factors — as introduced by Fama and French (1993). Carhart (1997) shows a four-factor model that includes a momentum factor performs superior to a three-factor model in explaining mutual fund performances. Jegadeesh and Titman (2001) show the momentum premium persists over the 1990–1998 sample period, which is after the sample included in Jegadeesh and Titman (1993).

To analyze the robustness of momentum, we next repeat the empirical exercise described in the previous section for (i) the post-JT sample period (1990 - 2024), and (ii) the pre-JT sample period that is available in the CRSP dataset (1927 - 1964).³ Table 1 contains the

³Note that we start in 1927 as the CRSP dataset offers data as of January 1926 and we need a 12 months burn-in period to construct the momentum signal.

results.

Focusing first on value-weighted portfolios, we observe that momentum is economically sizable and statistically significant also over the post-JT sample. Between 1990 and 2024 the winner-minus-loser portfolio returned on average 7.89% per annum with a t-statistic of 2.07, slightly below the results over the JT sample. For equally weighted portfolios, the momentum return spread was also sizable and significant (10.00%, t-statistic = 2.96). Similarly, we observe a sizable momentum premium over the pre-JT sample period. Between 1927 and 1964, average value-weighted momentum returns were 9.13% per annum with a t-statistic of 2.47, and equally weighted momentum returns were 7.14% with a t-statistic that is significant at the 10% significance level (t-statistic = 1.78). Overall, the momentum premium has been sizable and significant over the JT sample, the post-JT sample, and the pre-JT sample, averaging 9.24% (value weighted) or 9.75% (equally weighted) between 1927 and 2024.

Figure 2 shows the cumulative return of \$1 invested over these extended periods. We find that on average momentum returns were positive over both periods, but also experienced two significant ‘out-of-sample’ crashes — the first around the market reversal in 1929, the second around the market reversal in 2009. After these crashes, momentum returns have been strong again, with especially strong returns in the last years of the sample since the 2020s. In other words, momentum is persistent over these sample periods, but is also exposed to crash risk. More on this later in this article.

3.2 International evidence

Although the U.S. stock market represents a sizable chunk of the market capitalizations of global markets, most stocks are listed outside of the U.S.. Hence, another way to examine the robustness of stock-level momentum is to look at its evidence of momentum across international stock markets. Rouwenhorst (1998) research momentum internationally across twelve European countries between 1980 and 1995, finding persistent evidence of momentum also outside the U.S. stock market.

Next, we replicate and extend the international evidence for momentum. To this end we take global data from (Jensen et al., 2023) available via the JKP website and corresponding GitHub.⁴ As before, we construct value-weighted quintile portfolios, but do this per country to control for country-specific risks, which we aggregate to various regions using country market capitalizations.⁵ Data is available over the 1990 to 2024 period from a total of 46 countries. Panel a of figure 3 presents the results for momentum across key international regions: emerging markets (EM), as well as the worldwide developed and emerging markets including the U.S. (World) and excluding the U.S. (World-ExUS). For reference we also include the U.S. (US) results.

[Figure 3 about here]

The results show attractive momentum returns across markets. In worldwide markets (excluding the U.S.), we observe an average (value-weighted) momentum premium of 7.57% since 1990. A similar momentum premium is observed in emerging markets with 6.20%

⁴See jkpfactors.com]

⁵Note that this differs from data published on jkpfactors.com, which is based on tercile portfolios per country.

average annual returns. These numbers are comparable to the momentum returns in the U.S. over this sample period (of 7.89%). The resulting world momentum portfolio returned 7.77% on average. Moreover, all these numbers are significant at the 5% (U.S.) or 1% level (World-ExUS, EM, World). Figure 4 shows the cumulative returns of the long-short momentum portfolios, showing they are generally consistent across the various universes, but all displayed a significant momentum crash around the 2009 market reversal. Also recent returns have been strong, especially outside the U.S..

[Figure 4 about here]

The international dataset contains returns on all key listed stocks in 46 different countries. This allows for the opportunity to also test momentum per individual market.

In panel b of figure 3, we present the average annual long-short momentum returns within each country, sorted from markets with the highest momentum returns to the lowest. We only present results for the 31 countries that have data since 1990, but like to stress that the picture is similar when also including the 15 other countries with missing data. The results show positive momentum returns in all 31 countries, with Portugal leading the pack, and Japan having only slightly positive momentum returns. The right side of the figure shows the statistical significance in form of t-statistics of the long-short portfolios. The majority (19) of the 31 depicted markets have t-statistics above 1.96.

3.3 Deep sample evidence

As further test to the robustness of momentum we consider a unique sample of stocks over the period before CRSP was available: 1866-1926. Baltussen, van Vliet, and Van Vliet

(2021) created a database of U.S. stocks over this period based on the same source files that the CRSP database is build upon. Their dataset contains price and dividend data for 1,488 stocks that is augmented with hand-collected data on market capitalizations, allowing for a unique test on momentum factors over 61 years of out-of-sample data - a period in length exceeding the original JT sample. This ‘pre-CRSP’ period was a period of strong economic growth and rapid industrial developments, with stock markets playing a pivotal role. Consequently, conducting tests in the pre-CRSP era provides a powerful ground for independent out-of-sample tests. We utilize their dataset and construct quintile sorts on price momentum, in line with the empirical setup presented above.

Table 2 contains the results. Over the ‘pre-CRSP’ era stock momentum returns are again strong, averaging 8.18% per annum with a t-statistic of 2.77. This compares to 9.24% over the CRSP sample (1927-2024). Again the equally weighted numbers are similar, with 7.67% average return per annum (t-statistic = 2.57).

Using the pre-CRSP sample we can also construct the longest series of momentum returns to-date: 1866-2024. Over this 159 years of data, momentum returns have been strong and consistent, averaging 8.83% per annum (value-weighted) or 8.95% per annum (equally weighted) with t-statistics around 5. Figure 2 illustrates this by also including the pre-CRSP period. Since 1866 a \$1 invested in the momentum winner-minus-loser portfolio has grown to over \$10,000 - or a cumulative (gross) return of over 10,000% - for both the value-weighted and equally weighted portfolios.⁶

Overall, we can conclude that momentum shows up strongly across all major samples:

⁶We like to stress these returns are gross of transaction costs and management fees. Please see) for a discussion on momentum’s transaction costs, and Baltussen, van Vliet, and Van Vliet (2021) for transaction costs over the earlier samples.

the JT sample, the pre- and post-JT sample, across developed and emerging stock markets, and the pre-CRSP out-of-sample period since 1866. Further, as a wealth of studies show momentum is not only present in individual stock returns across the globe, but also in other asset classes like high-yield corporate bonds (Jostova, Nikolova, Philipov, and Stahel (2013)), equity futures, government bond markets, currencies, and commodities ((Asness et al., 2013)), a finding that holds persistently out-of-sample over a sample spanning the last 220 years, as shown by Baltussen, Swinkels, and Van Vliet (2021)). In other words, momentum is an *eternal* feature of financial markets.

3.4 Out-of-sample and post-publication decay of Momentum

McLean and Pontiff (2016) argue that performances of anomalies can decay because of post-discovery arbitrage. In fact, they find a post-publication performance decay of on average 58% across a wide set of stock-level anomalies over on average 13 years of post-publication data, which they attribute to arbitrage trading. This raises the question to which extent has the price momentum premium decayed? Next, we combine the datasets presented above to study this question formally.

A complicating factor for studying factor performance decay due to post-publication arbitrage is that also data mining implies out-of-sample decay. Interestingly, the period before the start date of the CRSP dataset (i.e., 1926) provides a unique opportunity, as it negates the arbitrage hypothesis for factor decay, since investors at that time could not have traded on insights from future research.

To study out-of-sample decay, we test the performance differences of the high-minus-low momentum portfolios over various periods using an OLS regression. The samples we consider

are: (i) the 1965-1989 JT sample period, or the 'in-sample' period, and dummy variables relative to this period for (ii) the 1990-2024 'post-publication period'⁷, (iii) the 1927-1964 'pre-sample CRSP' period, and (iv) the 1866-1926 'pre-CRSP' sample period. All these samples would be influenced by the effect of data mining, albeit in different degrees; the in-sample period the most, the pre-CRSP period the least as it has been hardly been used for testing before. In contrast, only the post-publication period is influenced by potential post-publication decay coming from arbitrage activity.

The results of this exercise are as follows. Returns spreads on the momentum high-minus-low portfolios are generally of similar size over the various samples. For value-weighted portfolios the average return in-sample is 11.28%, the combined coefficient for the three periods (reflecting an estimate of the impact of data mining) equals -2.15%, and the effect of post-publication decay equals -1.24%. Both coefficients are not significantly different from zero with t-statistics very close to zero (-0.21 and -0.24, respectively). Equally weighted portfolios reveal a slightly higher impact of data mining (-6.24%), but again this coefficient is statistically insignificant. In summary, we find consistent evidence for the momentum premium over time, and fail to find significant out-of-sample or post-publication decay in momentum performance.

3.5 Portfolio design choices

Above we have examined momentum using relatively standard portfolio design choices: using quintile and either value-weighting or equal-weighting. However, several studies argue

⁷McLean and Pontiff (2016) argue there is a difference in decay post the initial sample period but before publication, and post-publication. We have verified that changing the period to 1994-2024 does not change conclusions.

portfolio design choices can materially influenced observed factor returns. Menkveld et al. (2024) show that degrees of freedom in research design choices can lead to major variation in outcomes, resulting in “non-standard errors”. Soebhag, Van Vliet, and Verwijmeren (2024) show that non-standard errors are substantial as portfolio design choices materially affect factor returns and model selection exercises. They recommend researchers to conduct specification checks to mitigate concerns that results are driven by selecting a particular set of choices. Hence, next we consider a broad range of portfolio design choices.

We construct thousands of different versions of the momentum factor in our sample across a set of 12 factor construction choices, building on the construction choices used in Soebhag et al. (2024), expanded with a transaction-cost-mitigation technique from Novy-Marx and Velikov (2019). Specifically, we focus on 2x3 size-momentum double-sorted portfolios, an approach followed by Fama and French (1993) and many others to control for size risk exposures in factors, thereby adding robustness to the single-sorting approach followed above. We alternate between the following decisions: (i) 30-70 breakpoints versus 20-80, (ii) NYSE versus full-sample breakpoints, (iii) including versus excluding financial firms, (iv) applying industry neutralization or not, (v) independent versus dependent sorts, (vi) monthly rebalancing or yearly (at the end of June), (vii) value-weighting returns or equal-weighting, (viii) including or excluding micro-caps, (ix) remove or keep stocks with negative book-to-market ratios, (x) remove or keep stocks with prices below 5 dollars in the sorting month, (xi) include or exclude utility stocks, and (xii) banding portfolio returns and transaction costs.⁸ Combining these 12 choices results in 4,096 unique specifications, and thereby different versions of factor returns. For each, we compute the returns and run our model selection exercises.

⁸All factors are constructed via 2-by-3 sorting on size and a factor characteristic.

For this analysis, we stick to the sample period of 1972 - 2023 used by Soebhag et al. (2024).

Figure 5 shows the resulting empirical distribution of (annualized) Sharpe ratios for the winner-minus-loser momentum factor based on the 4,096 distinct factor construction specifications. The momentum factor exhibits an empirical Sharpe ratio distribution that is well-positive across all 4,096 design choices. The median Sharpe ratio equals 0.61, varying between 0.38 and 0.94.⁹ Over the sample of 52 years of data, all these Sharpe ratios are statistically highly significant with t-statistics varying between 2.74 and 6.78. Overall, the Sharpe ratio distribution reveals the momentum factor shows strong and consistent risk-adjusted returns across thousands of factor construction choices.

4 Momentum is much more than Price Momentum

Price momentum is an eternal feature of equity markets. In this section, we examine the evolution of momentum beyond purely price-based measures and present their empirical evidence.

4.1 The evolution of Momentum

Figure 1 — showing the (albeit incomplete and rough) timeline of various studies to the momentum effect — highlights that after the initial discovery of price momentum, studies since the mid-1990s have turned to examining robustness but also other manifestations of momentum in stock markets.

⁹Note that the average volatility of the 2x3-sorted momentum factor is 14.20% per annum, and hence these numbers compare to an average long-short return of 8.87% per annum, similar to the numbers present in the previous tables and figures.

Chan, Jegadeesh, and Lakonishok (1996) shows momentum can also be observed in fundamentals like earnings predicting stock returns. Moskowitz and Grinblatt (1999) find momentum is also present in industry returns. Subsequent studies highlight how stock momentum returns can be measured using unexpected earnings and analyst recommendations (Chordia & Shivakumar, 2006), firm-specific returns (Blitz, Huij, & Martens, 2011), momentum in news (Wang, Zhang, & Zhu, 2018), momentum in connected firms based on alternative industry networks (Hoberg & Phillips, 2018) and analyst networks (Ali & Hirshleifer, 2020), momentum in factor returns itself (Gupta & Kelly, 2018; Arnott, Kalesnik, & Linnainmaa, 2023), and momentum based on anchors (Büsing, Mohrschladt, & Siedhoff, 2024). All these measures allow for the measurement of momentum beyond what is observed in prices of the stock itself.

Next, we empirically examine the effectiveness of these measures in predicting future stock returns, examine their overlap with traditional price momentum, and compare individual (i.e., ‘one-dimensional’) versus ‘multi-dimensional’ approaches of momentum.

4.2 Fundamental Momentum

The prior literature documents that not only stock prices exhibit a momentum effect, but momentum is observable in stock fundamentals as well. In particular, Chordia and Shivakumar (2006) and Jegadeesh and Livnat (2006) show momentum in earnings measured by the Standardized Unexpected Earnings (‘SUE’). Furthermore, Chan et al. (1996) find momentum in analyst forecast revisions using the previous six months in analyst revisions for a stock (‘REV6’) as a proxy. Lastly, Wang et al. (2018) present evidence for momentum in firm news (‘NEWS’), i.e., positive news about a firm in the past generates positive news in

the future, predicting future stock returns.

We empirically examine these fundamental momentum definitions in the U.S. stock markets using the CRSP sample and methodology outlined in the previous sections. The SUE measure from Jegadeesh and Livnat (2006) and the REV6 measure from Chan et al. (1996) are constructed using data from Compustat and IBES, respectively. For news momentum, we use the composite news sentiment scores from Ravenpack on Dow Jones Newswires articles for the same news categories as used by Wang et al. (2018).¹⁰ After constructing these variables, we form quintile portfolios, as before, and examine their future return.

Table 3, Panel A, contains the results. We consider two samples; (i) the longest sample over which we can construct the measures, and (ii) if possible the post JT sample (1990-2024) as utilized in previous tables. The equally weighted long-short portfolios consistently yield both economically and statistically significant returns for all three measures, with alpha ranging between 4.22% for NEWS and 8.61% for SUE. The value-weighted sorts yield more mixed results. SUE has a return spread of 2.58% from 1965 to 2024 is accompanied by a t-statistic of 2.41, which decayed to an insignificant 1.15% over the post JT sample. REV6 is significant in both samples, including the sample starting from 1990 with an average return of 5.48% and a t-statistic of 2.73. News momentum is insignificant with a value-weighted long-short spread of 1.15% per annum. Hence, fundamental momentum is most robustly present in analyst recommendations and stronger in smaller companies that receive more weight in equally weighted portfolios.

We also combine the above three measures in an equally weighted fundamental momentum composite ('EW_FUND'). Long-short returns on this composite are again sizable and

¹⁰The Ravenpack data is available since 2000.

highly statistically significant. Overall, we conclude that momentum manifests not only in stock prices but also in firm fundamentals.

[Table 3 about here]

4.3 Firm-specific Momentum

Traditional price momentum strategies rely on total stock returns. Among others, Blitz et al. (2011) show that high exposures of stock returns to systematic risk factors — like market beta, size, and value — may lead to excessively high return variability in price momentum strategies. To remedy this, they propose a measure that measures momentum in the residual, or firm-specific, returns relative to the Fama-French 3-factor model. They find that this residual momentum ('RES_MOM') strategy leads to a sizable reduction in portfolio return volatility, and to higher risk-adjusted returns.

We next empirically examine the residual momentum measure of Blitz et al. (2011) using each stock's residual returns relative to the Fama-French three factor model, following (Jensen et al., 2023). Panel B of table 3 present the results. For both value-weighted and equally weighted long-short portfolios, RES_MOM has sizable return spreads that are all statistically significant. For example, over the 1965–2024 period, the average return is 7.56% with a t-statistic of 5.50. Interestingly, t-statistics exceed those of traditional price momentum (e.g., over the post JT sample, value-weighted: 3.45 vs 2.07, and equally weighted: 6.54 vs 2.96), as the volatility of the portfolios is materially reduced. Overall, we conclude that momentum is also significantly present in firm-specific returns.

4.4 Anchor-based Momentum

Traditional momentum strategies rely on past returns over fixed intervals (e.g., 6 or 12 months). George and Hwang (2004) ask whether an even more straightforward and easily accessible indicator - nearness of a stock’s current price to its 52-week high - could better predict future returns. The authors propose that investors use the 52-week high as an *anchor* when assessing the impact of new information. Stocks near their high may see underreaction — investors hesitate to push prices even higher despite positive news; stocks far from their high may not fall enough despite bad news. This anchoring creates momentum-like effects.

Interestingly, using the 52-week high price, Büsing et al. (2024) show that price momentum can be split in two parts: $MOM = \frac{P_1}{P_0} = \frac{P_{high}}{P_0} \frac{P_1}{P_{high}} = HTP \times PTH$, where P_0 is the stock price at the beginning of month $t - 12$ and P_1 is the stock price at the end of month $t - 1$. PTH denotes the price-to-high ratio, as used by George and Hwang (2004), and HTP refers to the high-to-price ratio. Büsing et al. (2024) find that especially the HTP component contributes to over 80% of overall price momentum profits.

We also empirically construct both the PTH and HTP measures, as presented in Panel C of table 3. In line with Büsing et al. (2024), momentum profits are mainly driven by the HTP component rather than the PTH component. PTH long-short return spreads are small and insignificant for value-weighted portfolios (0.17% since 1927 with a t-statistic of 0.08) and equally weighted portfolios (3.52% with a t-statistic of 1.42). In contrast, HTP long-short return spreads are all sizable and significant across samples. For example, value-weighted long-short returns average 9.05% since 1927 with a t-statistic of 5.35. We also combine the above two measures in an equally weighted anchor-based momentum composite ('EW_ANCHOR'). Long-short returns on this composite are sizable and highly statistically

significant. Overall, we conclude that also momentum measured based on observable anchors delivers strong premiums.

4.5 Industry and Network Momentum

Another dimension in which momentum evolved is to measure returns based on peers, such as based on industries and networks. (Moskowitz & Grinblatt, 1999) introduce industry momentum strategies that take long positions in stocks from past winning industries and short positions in stocks from past losing industries. Building on industry momentum, (Hoberg & Phillips, 2018) aim to refine traditional industry classifications, such as SIC codes, with text-based industry classifications ('TNIC') obtained from 10K-filings. These TNIC peer groups circumvent the assumption of traditional industry classifiers that each firm can only be part of one industry. At the same time, it allows for measuring momentum in the returns of less visible firms connected to a firm, or network momentum. Rather than looking at the relatedness of stocks based on their industries, (Ali & Hirshleifer, 2020) investigate stock networks constructed using shared analyst coverage. One motivation behind this approach is that analysts are more likely to cover firms with linked fundamentals.

Our empirical application next implements these three alternative momentum strategies. First, we use the Fama-French 49 industries to compute classical industry momentum ('IND_MOM') on a stock level using CRSP industry data. Industry returns for a stock are defined as the value-weighted peer stock returns in the same industry, excluding the stock itself. Following Moskowitz and Grinblatt (1999), industry momentum does not exclude the returns of the prior month and hence includes the returns from months $t-12$ to t . Second, for the text-based industries of Hoberg and Phillips (2018), we employ the Embeddings-based

TNIC ('ETNIC') peer similarity scores introduced by Hoberg and Phillips (2025) that are available as of 2000. We select the fifty stocks with the highest similarity scores to compute the ETNIC momentum returns for each stock. Third, we construct connected analyst momentum ('C_Anly'). Here, we assume a stock to be covered by an analyst if the analyst has published at least one FY1 or FY2 earnings forecast over the past 12 months. The returns are then weighted based on the number of shared analysts for each stock.

Panel D of table 3 shows the results. For the equally weighted long-short portfolios, all strategies are statistically significant with average returns exceeding 10% and t-statistics between 2.53 for ETNIC to 8.16 for IND_MOM. As for fundamental and anchor-based momentum, value-weighted long-short returns are on average lower but sizable and generally significant. IND_MOM long-short returns average to 6.73% with a t-statistic of 4.52 since 1927, ETNIC long-short returns average to 9.79% (t-statistic = 1.93) since 2000, and C_Anly spread return average to 7.55% (t-statistic = 2.65) since 1983. We also combine the above three measures in an equally weighted network momentum composite ('EW_NETWORK'). Long-short returns on this composite are again sizable and highly statistically significant. Overall, we conclude that momentum can also be measured in a firm's network.

4.6 Factor Momentum

The final evolution in momentum that we consider is factor momentum. Gupta and Kelly (2018), Ehsani and Linnainmaa (2022), and Arnott et al. (2023) find that factors with positive returns in the previous year tend to continue earning positive premiums the next months, or what they term factor momentum. Moreover, these studies argue that it captures part of the traditional price momentum effect. This means that another manner to measure

momentum at the stock-level is by a stock’s exposure to factors times past factor returns.

Next, we also recreate factor momentum empirically using data of 153 risk factors from Jensen et al. (2023). In line with Ehsani and Linnainmaa (2022), we define factor momentum as the past 12 months of factor returns (hence not skipping the most recent month), and create quintile portfolios as before. The results are presented in Panel E of table 3. As for the other momentum measures defined above, also factor momentum returns are sizable and significant. The average return on the value-weighted long-short strategy equals 6.45% since 1927 with a t-statistic of 3.52, while equally weighted portfolio returns are about double. Average return spreads are similar over the more recent post JT sample. Overall, we conclude that momentum is also significantly present in factors.

4.7 Momentum is multi-dimensional

The above results show momentum is present in prices, fundamentals, firm-specific returns, anchor-based returns, returns measured from networks, and returns measured from factors. This triggers the questions of how these measures interact and to which extent they complement each other.

To analyze this, we first analyze correlations between the various momentum long-short strategy returns. Figure 6 presents the results. The correlations between all alternative momentum returns and the traditional price momentum returns are positive and generally high, for example varying between 0.17 for C_ANLY and 0.77 for IND_MOM for value-weighted portfolios.

[Figure 6 about here]

Next, we perform spanning regressions in which we regress the returns of these alternative momentum strategies on an intercept and the traditional price momentum returns on the longest sample available. Table 4 shows the results. In line with the correlation results, each of the alternative momentum strategies load positively on price momentum, with all slope coefficients being highly statistically significant. Further, most spanning alphas are positive and statistically significant, especially for equally weighted portfolios, thereby indicating added value over traditional price momentum strategies.

[Table 4 about here]

Focusing on the composites constructed in the previous sub-sections, we observe positive and significant value-weighted portfolio spanning alphas for fundamental momentum, firm-specific momentum, and network momentum. The anchor-based momentum composite is insignificant as HTP dominates PTH, while factor momentum is dominated by price momentum when value weighting. However, for the equally weighted portfolio the spanning alphas of all composites are positive and significant. In other words, factor momentum adds value to price momentum when giving more weight to smaller stocks.

Finally, we construct an (equally weighted) momentum composite across all momentum measurements analyzed above ('EW_ALL'). The results of this 'multi-dimensional' momentum approach are presented in Panel F of table 3, and the final rows of figure 6 and table 4. Figure 7 depicts the cumulative returns of the traditional price momentum and multi-dimensional momentum strategies since 1927. First, we observe that the value-weighted long-short returns is sizable (6.35% per annum) and highly significant (t-statistic = 5.43). The t-statistics exceed those of traditional price momentum, suggesting a superior risk-return trade off for the multi-dimensional momentum strategy. Second, the spanning alpha relative

to traditional price momentum is positive (1.76% per annum) and significant (t-statistic = 3.63). Hence, the multi-dimensional momentum strategy cannot be explained by traditional price momentum. When turning to equally weighting these numbers further increase (average return = 9.65% with a t-statistic of 8.22, spanning alpha = 4.83% with a t-statistic of 8.61). In other words, the added value of the multi-dimensional momentum strategy becomes larger when weighting more to smaller stocks. Overall, we conclude that momentum has evolved from traditional price momentum to a multi-dimensional approach embedding trends beyond prices.

[Figure 7 about here]

5 Momentum and its risks

Several studies show momentum is exposed to crash risk (Barroso & Santa-Clara, 2015; Daniel & Moskowitz, 2016), as also became evident in the previous sections from figures 2, 4, and 7. In this section, we evaluate the risk profile of momentum strategies, including the fundamental, firm-specific, anchor-based, network, factor, and multi-dimensional momentum approaches. Further, we examine crash-managed momentum strategies designed to significantly enhance the risk-efficiency of the momentum factor.

5.1 Momentum has crash risk

To examine momentum crashes, we examine the distribution of returns of the various momentum return series. Table 5 shows the results in terms of return volatility ('Vol'), skewness ('Skew'), kurtosis ('Kurt'), and maximum drawdown ('MD') in momentum's cumulative re-

turn. The returns of traditional price momentum are left-skewed, display excess kurtosis, and are exposed to significant crashes with a maximum drawdown of -88.41% for the value-weighted portfolios over the 1927–2024 sample. Figure 2 revealed a similar crash profile also over the pre-CRSP sample, suggesting crash risk is a persistent feature of traditional price momentum strategies.

[Table 5 about here]

5.2 Reducing Momentum’s crash risk

As for the alternative momentum strategies, the risk, as measured by the volatility and maximum drawdown is generally lower than for standard price momentum. A clear exception to this rule is the PTH ratio achieving a higher maximum drawdown, whereas HTP actually improves upon price momentum when it comes to the volatility and the drawdowns. This finding suggest PTH is a sizable driver of momentum’s crash risk. The equal-weighted combination of all momentum strategies is also shown to be less risky than standard price momentum in terms of volatility and maximum drawdowns. Overall, these results confirm that crashes may be an inherent feature of the momentum trade, but at the same time can be mitigated by the alternative momentum measures.

Finally, Barroso and Santa-Clara (2015) show that limiting exposure to crash risk substantially improves momentum returns. As our final exercise, we repeat their approach for traditional price momentum. Using daily returns, they scale the momentum returns using the realized variance over the previous 126 days. In a similar vein, we scale our long-short

momentum returns using daily momentum returns data from Jensen et al. (2023).¹¹ Additionally, we perform the scaling on a stock level by scaling the momentum returns for a stock using daily stock returns data. We refer to the former risk-managed momentum strategy as ‘RM_MOM’ and to the latter as ‘RM_MOM_SL’.

The final two rows of table 5 contain the results of these risk-managed momentum strategies. Both of these strategies show a clear reduction of the maximum drawdown from -88.41% to -49.37 and -54.64%. RM_MOM also shows a considerable increase in returns compared to standard price momentum from 9.24% to 17.87% with similar volatilities around 20%. Although the stock-level risk-managed momentum strategy shows a similar average return of 9.36%, there is a sizable reduction in volatility to 15.33%. Volatility-scaled momentum thus has a materially higher Sharpe ratio and a more normal return distribution. Overall, we conclude that momentum crash risk can be mitigated by accounting for alternative momentum measures, risk-aware momentum measurement, or scaling momentum by volatility, all making momentum portfolios more risk-efficient.

6 What drives the Momentum premium?

Having established the empirical evidence for momentum, a natural question is what is the economic driver of the returns on the momentum factor. Although a comprehensive study into the explanations of momentum is beyond the scope of this paper, several important observations based on the results presented in this study help understand momentum.

First, momentum is persistent across time and markets, including 61 years of unique out-

¹¹Note that the momentum factor returns of Jensen et al. (2023) are long-short tercile portfolios, whereas our monthly momentum returns are based on quintile sorts.

of-sample data. This makes an explanation based on data mining highly unlikely, leading us to believe that stock-level momentum is an eternal feature of financial markets. The level of the premium is also sizable, suggesting a high-level of economic significance. Although we do not study the impact of transaction costs in this study, work from academics and practitioners suggest these conclusions also hold after transaction costs (Novy-Marx & Velikov, 2016; Israel & Moskowitz, 2013).

Second, momentum strategies might earn excess returns as a compensation for bearing certain risks. Examples include crash risk, macroeconomic and liquidity risk, or downside risks. For example, Liu and Zhang (2008) links momentum to investment-based risk, while Sadka (2006) and Pástor and Stambaugh (2003) argue that liquidity risk explains part of variation in momentum profits. However, both Asness et al. (2013) and Baltussen, van Vliet, and Van Vliet (2021) show that although these explanations have some connection to the momentum premium, their impact is generally small. Kelly, Moskowitz, and Pruitt (2021) argue that momentum portfolios exhibit higher conditional market betas during adverse states, implying compensation for downside risk. However, Baltussen, van Vliet, and Van Vliet (2021) show that downside risk cannot explain the momentum premium across 154 years of risk events. Further, the evidence presented in this paper shows persistent momentum returns that become stronger when mitigating its crash risk. Overall, while these risk-based explanations offer partial insights, empirical evidence suggests they fail to explain the momentum factor premium.

Instead, a behavioral explanation seems more likely driven by investor underreaction or overreaction. In fact, several studies highlighted in figure 1 argue behavioral bias combined with limits to arbitrage drive momentum. We close this paper on momentum with discussing

seminal behavioral papers on the momentum factor.

Barberis, Shleifer, and Vishny (1998) argue that investors tend to incorporate information gradually. New information leads to delayed price adjustments. When firms release news, positive or negative, prices may not fully reflect this news right away, as investors anchor to prior expectations or update beliefs too conservatively. The slow diffusion of information means that prices continue to drift in the direction of the news, creating a continuation of prices.

Daniel, Hirshleifer, and Subrahmanyam (1998) propose that investors exhibit overconfidence in the precision of their private signals. Investors credit themselves when signals are confirmed by outcomes, but blame external factors when signals fail. This leads to overreaction, where prices can overshoot in the short run, reinforcing price continuation. This overshoot can be even higher when investors herd around perceived signals.

Hong and Stein (1999) propose that gradual information diffusion creates momentum in the short to medium term and reversals in the long run. Momentum arises because “newswatchers” underreact to firm-specific information, while “momentum traders” amplify trends.

Grinblatt and Han (2005) show that investors are reluctant to sell assets at a loss relative to their purchase price. This disposition effect causes slow selling of losing stocks and quicker selling of winners. Prices therefore deviate from fundamentals in a predictable way, producing continuation patterns consistent with momentum.

7 Conclusion

In conclusion, stock momentum has firmly established itself as a foundational factor in equity markets, supported by robust and consistent empirical evidence across extensive time frames (spanning up to 159 years of data), diverse portfolio designs (including over 4,000 different specifications), and global stock markets (from smaller emerging economies to larger developed markets). What began as a simple price-based strategy has evolved into a multi-dimensional phenomenon that incorporates fundamental, firm-specific, anchor-based, network, and factor-based signals, enhancing the return–risk trade-off of momentum. Moreover, momentum strategies have demonstrated resilience against concerns of data mining and arbitrage, with little evidence of performance decay driven by arbitrage pressures. Advances in multi-dimensional and risk-managed approaches have also significantly reduced the inherent crash risk historically associated with momentum investing, further solidifying its role as a systematic investment style. Ultimately, momentum is an eternal feature of financial markets — and hence important for both investors and academics to consider.

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A Tables and Figures

Table 1: **Price Momentum in the US.** This table displays the average annualized returns (in %) of quintile price Momentum portfolio sorts using US stocks. T-statistics are shown in parentheses. We show the Value-Weighted (VW) and Equally-Weighted (EW) returns over various sample periods. Panel A shows the results of the sample used by (Jegadeesh & Titman, 1993), denoted as the JT sample (1965-1989). Panel B shows the results for the Pre-JT sample (1927-1964), the Post-JT (1990-2024) sample, and the full sample period (1927-2024).

Panel A: JT Sample							
Returns	Sample	1	2	3	4	5	5-1
VW	1965-1989	6.88 (1.70)	9.59 (2.97)	10.38 (3.21)	13.41 (3.97)	18.16 (4.50)	11.28 (3.50)
EW	1965-1989	9.05 (1.85)	14.06 (3.62)	16.06 (4.33)	18.12 (4.74)	22.43 (4.96)	13.38 (4.78)
Panel B: Post-, Pre-, and Full-Sample							
Returns	Sample	1	2	3	4	5	5-1
VW	1990-2024	7.18 (1.72)	10.78 (3.75)	10.53 (4.38)	11.66 (4.77)	15.07 (4.50)	7.89 (2.07)
	1927-1964	7.38 (1.34)	9.74 (2.23)	11.94 (3.11)	13.94 (3.97)	16.51 (4.53)	9.13 (2.47)
	1927-2024	7.18 (2.56)	10.07 (4.69)	11.04 (5.78)	12.99 (7.09)	16.42 (7.75)	9.24 (4.31)
	1990-2024	7.69 (1.58)	11.89 (4.01)	13.26 (5.06)	15.28 (5.57)	17.69 (4.66)	10.00 (2.96)
EW	1990-2024	7.69 (1.58)	11.89 (4.01)	13.26 (5.06)	15.28 (5.57)	17.69 (4.66)	10.00 (2.96)
	1927-1964	13.56 (2.05)	14.96 (2.80)	15.75 (3.39)	16.79 (3.91)	20.70 (4.85)	7.14 (1.78)
	1927-2024	10.31 (3.09)	13.64 (5.39)	14.94 (6.67)	16.59 (7.66)	20.06 (8.26)	9.75 (4.66)
	1990-2024	7.69 (1.58)	11.89 (4.01)	13.26 (5.06)	15.28 (5.57)	17.69 (4.66)	10.00 (2.96)

Table 2: **Price Momentum in the US in the pre-CRSP sample.** This table displays the average annualized returns (in %) of quintile price Momentum portfolio sorts using monthly US stock data going back to 1866. T-statistics are shown in parentheses. We show the Value-Weighted (VW) and Equally-Weighted (EW) returns.

Returns	Sample	1	2	3	4	5	5-1
VW	1866-1926	5.36	6.24	9.43	9.61	13.54	8.18
		(1.62)	(3.20)	(6.58)	(6.59)	(6.23)	(2.77)
	1866-2024	6.48	8.60	10.42	11.69	15.32	8.83
		(3.02)	(5.66)	(8.02)	(9.28)	(9.88)	(5.08)
EW	1866-1926	7.11	6.29	9.21	11.03	14.79	7.67
		(2.02)	(3.04)	(6.32)	(7.07)	(6.50)	(2.57)
	1866-2024	9.09	10.82	12.74	14.46	18.04	8.95
		(3.68)	(6.18)	(8.55)	(9.87)	(10.40)	(5.19)

Table 3: **Alternative momentum strategies results.** This table shows the annualized returns (in %) of quintile portfolios based on alternative momentum measures. T-statistics are shown in parentheses. The portfolio sorts are Value-Weighted (VW) or Equal-Weighted (EW), and are based on US data. Panel A includes portfolio sorts on momentum in stock fundamentals: standardized unexpected earnings (SUE), six-month rolling analyst revisions (REV6), and news sentiment (NEWS). EW_FUND is the EW average of the three fundamental momentum strategies. Panel B shows firm-specific momentum with residual momentum (RES_MOM) computed using the three Fama-French factors. The anchor-based momentum results in Panel C include the price-to-high (PTH) and high-to-price (HTP) ratios. EW_ANCHOR is the EW average of these anchor-based momentum strategies. Panel D includes industry momentum using the FF-49 industries (IND_MOM) and the embeddings-based text-based industry classification (ETNIC). This panel also includes Network Momentum in the form of shared analyst coverage (C_Anly). EW_NETWORK is the EW average of the three network-based momentum strategies. Finally, Panel E shows the factor momentum (F_MOM) results and Panel F shows the equal-weighted combination of all alternative momentum strategies presented in this table, including standard US price momentum.

Panel A: Fundamental Momentum													
		VW						EW					
Char	Sample	1	2	3	4	5	5-1	1	2	3	4	5	5-1
SUE	1965-2024	10.04 (4.66)	8.96 (4.40)	11.67 (5.77)	12.45 (6.15)	12.62 (5.92)	2.58 (2.41)	8.17 (2.98)	12.51 (4.80)	14.29 (5.50)	15.93 (6.15)	16.77 (6.41)	8.61 (10.94)
	1990-2024	10.54 (3.66)	9.31 (3.53)	11.50 (4.52)	13.08 (5.10)	11.69 (4.30)	1.15 (0.86)	8.11 (2.24)	12.28 (3.65)	13.66 (4.09)	15.50 (4.58)	15.78 (4.55)	7.67 (7.96)
REV6	1983-2024	9.49 (3.09)	10.97 (4.31)	11.76 (5.19)	13.25 (5.69)	14.96 (5.39)	5.46 (3.19)	6.96 (1.75)	11.70 (3.88)	12.57 (4.61)	13.60 (4.87)	15.27 (4.37)	8.31 (6.32)
	1990-2024	8.43 (2.45)	10.17 (3.62)	10.75 (4.39)	11.87 (4.70)	13.91 (4.59)	5.48 (2.73)	7.10 (1.56)	11.39 (3.41)	11.97 (4.01)	12.43 (4.07)	14.69 (3.74)	7.59 (4.96)
NEWS	2000-2023	8.44 (2.13)	1.34 (0.36)	10.03 (3.28)	7.96 (2.46)	9.58 (3.07)	1.15 (0.64)	8.38 (1.51)	7.87 (1.62)	11.92 (2.94)	10.30 (2.48)	12.60 (3.04)	4.22 (2.13)
EW_FUND	1965-2024	9.46	9.00	11.46	12.12	13.34	3.88	7.98	12.29	13.77	15.16	16.61	8.63
	1965-2024	(4.22)	(4.39)	(5.80)	(6.08)	(6.24)	(3.78)	(2.69)	(4.73)	(5.49)	(6.07)	(6.25)	(10.34)

(Table continues on next page.)

Table 3 (continued)

Panel B: Firm-specific Momentum													
Char	Sample	VW						EW					
		1	2	3	4	5	5-1	1	2	3	4	5	5-1
RES_MOM	1965-2024	7.82	10.13	11.17	12.60	15.38	7.56	7.54	11.22	14.14	16.98	20.91	13.38
		(3.45)	(4.92)	(5.44)	(6.37)	(7.30)	(5.50)	(2.61)	(4.22)	(5.48)	(6.49)	(7.94)	(10.71)
	1990-2024	7.99	10.78	11.13	12.64	14.65	6.66	7.04	10.40	12.88	16.20	19.48	12.44
		(2.63)	(4.06)	(4.19)	(5.04)	(5.50)	(3.45)	(1.81)	(3.03)	(3.93)	(4.90)	(5.96)	(6.54)
Panel C: Anchor-based Momentum													
Char	Sample	VW						EW					
		1	2	3	4	5	5-1	1	2	3	4	5	5-1
PTH	1927-2024	11.68	11.32	12.14	12.01	11.85	0.17	12.12	14.22	15.82	15.50	15.64	3.52
		(3.90)	(4.59)	(5.69)	(6.55)	(7.41)	(0.08)	(3.37)	(5.16)	(6.82)	(7.57)	(8.91)	(1.42)
	1990-2024	8.85	10.80	12.28	11.86	11.83	2.98	8.19	12.57	14.63	15.26	15.86	7.67
		(2.14)	(3.22)	(4.51)	(5.07)	(5.23)	(0.95)	(1.59)	(3.74)	(5.08)	(5.91)	(6.68)	(1.94)
HTP	1927-2024	7.45	10.25	12.01	14.43	16.50	9.05	8.61	12.12	14.41	16.20	18.58	9.97
		(4.14)	(5.54)	(6.20)	(6.49)	(6.35)	(5.35)	(3.68)	(5.31)	(5.83)	(6.17)	(6.02)	(6.63)
	1990-2024	7.39	10.45	11.24	12.78	15.94	8.55	7.52	11.47	13.18	14.58	15.44	7.92
		(2.72)	(4.09)	(4.33)	(4.49)	(4.09)	(2.69)	(2.24)	(3.71)	(4.15)	(4.23)	(3.45)	(3.11)
EW_ANCHOR	1927-2024	9.57	10.78	12.07	13.25	14.18	4.61	10.37	13.17	15.12	15.88	17.11	6.75
		(4.11)	(5.08)	(6.00)	(6.63)	(6.96)	(4.01)	(3.54)	(5.26)	(6.34)	(6.86)	(7.26)	(5.48)

(Table continues on next page.)

Table 3 (continued)

Panel D: Industry & Network Momentum													
Char	Sample	VW						EW					
		1	2	3	4	5	5-1	1	2	3	4	5	5-1
IND_MOM	1927-2024	8.17 (3.77)	9.42 (4.66)	12.15 (6.31)	13.67 (7.05)	14.90 (7.32)	6.73 (4.52)	8.90 (3.26)	12.28 (4.80)	15.42 (5.92)	16.70 (6.62)	20.16 (7.71)	11.26 (8.16)
	1990-2024	8.29 (2.59)	9.59 (3.43)	11.84 (4.64)	13.31 (5.06)	13.55 (4.66)	5.27 (1.93)	6.12 (1.55)	10.82 (3.24)	12.87 (3.98)	14.38 (4.18)	19.66 (5.00)	13.54 (4.89)
ETNIC	2000-2023	2.87 (0.47)	4.34 (0.78)	5.34 (0.99)	10.71 (2.06)	12.67 (2.53)	9.79 (1.93)	2.37 (0.34)	4.25 (0.72)	6.64 (1.19)	9.59 (1.77)	15.97 (2.98)	13.59 (2.53)
C_ONLY	1983-2024	7.02 (2.12)	10.91 (4.16)	13.38 (5.64)	14.83 (6.06)	14.57 (5.32)	7.55 (2.65)	2.75 (0.71)	8.89 (2.77)	12.67 (4.26)	16.38 (5.43)	19.34 (5.62)	16.58 (6.59)
	1990-2024	6.29 (1.69)	9.99 (3.47)	12.39 (4.81)	13.82 (5.26)	13.36 (4.42)	7.07 (2.14)	2.11 (0.48)	8.26 (2.30)	11.99 (3.63)	15.98 (4.79)	19.67 (5.06)	17.56 (5.96)
EW_NETWORK	1927-2024	7.27 (3.25)	9.17 (4.44)	12.04 (6.09)	14.15 (7.11)	15.41 (7.50)	8.13 (5.63)	8.03 (2.86)	11.50 (4.41)	15.00 (5.69)	17.06 (6.72)	20.29 (7.82)	12.26 (9.17)
Panel E: Factor Momentum													
Char	Sample	VW						EW					
		1	2	3	4	5	5-1	1	2	3	4	5	5-1
F_MOM	1927-2024	-1.41 (-1.47)	0.44 (0.65)	1.71 (3.02)	2.66 (3.69)	5.04 (4.76)	6.45 (3.52)	-2.59 (-2.20)	1.03 (1.15)	3.60 (5.88)	5.97 (7.68)	9.54 (7.66)	12.12 (5.50)
	1990-2024	-1.13 (-0.67)	0.67 (0.82)	1.70 (2.87)	2.77 (3.00)	4.57 (2.90)	5.69 (1.92)	-1.24 (-0.67)	2.12 (2.43)	4.63 (7.41)	6.47 (6.18)	9.47 (5.31)	10.71 (3.22)
Panel F: Equal-weighted Momentum													
Char	Sample	VW						EW					
		1	2	3	4	5	5-1	1	2	3	4	5	5-1
EW_ALL	1927-2024	7.13 (3.70)	8.59 (5.16)	10.31 (6.57)	11.89 (7.60)	13.47 (8.06)	6.35 (5.43)	7.73 (3.22)	10.87 (5.38)	13.13 (6.72)	14.75 (7.63)	17.38 (8.51)	9.65 (8.22)

Table 4: **Spanning regression results.** This table shows the results of spanning regressions of the long-short quintile portfolios with momentum-based measures as dependent variables. The spanning regressions regress the portfolio returns on an intercept and the returns of the long-short price momentum strategy based on US data. We include results for the VW momentum measures on VW price momentum, and for EW momentum measures on EW price momentum. Equal-weighted momentum strategy combinations are also included.

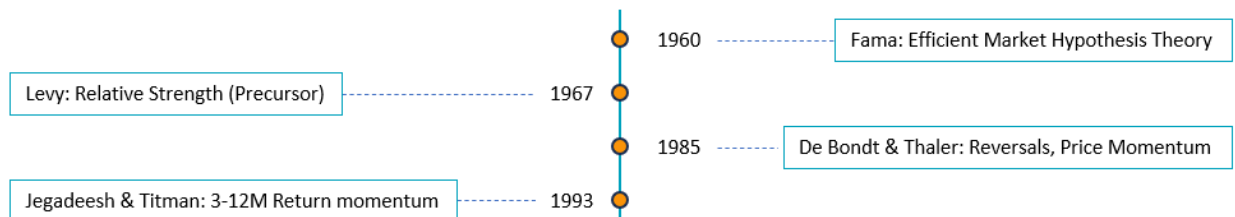
Char	Sample	VW				RSq
		Alpha		Beta		
SUE	1965-2024	0.71	(0.75)	0.20	(15.06)	0.24
REV6	1983-2024	3.86	(2.47)	0.22	(10.48)	0.18
NEWS	2000-2023	0.48	(0.33)	0.21	(12.20)	0.34
EW_FUND	1965-2024	1.85	(2.14)	0.22	(17.77)	0.31
RES_MOM	1965-2024	4.47	(4.14)	0.33	(21.70)	0.40
PTH	1927-2024	-5.43	(-3.50)	0.61	(28.98)	0.42
HTP	1927-2024	5.98	(3.86)	0.33	(15.90)	0.18
EW_ANCHOR	1927-2024	0.27	(0.49)	0.47	(61.98)	0.77
IND_MOM	1927-2024	1.82	(1.88)	0.53	(40.90)	0.59
ETNIC	2000-2023	7.66	(1.94)	0.64	(13.79)	0.40
C_ANLY	1983-2024	6.48	(2.29)	0.15	(3.85)	0.03
EW_NETWORK	1927-2024	3.82	(3.64)	0.47	(32.94)	0.48
F_MOM	1927-2024	1.88	(1.25)	0.49	(24.29)	0.33
EW_ALL	1927-2024	1.76	(3.63)	0.50	(75.89)	0.83

Char	Sample	EW				RSq
		Alpha		Beta		
SUE	1965-2024	6.51	(9.61)	0.18	(16.92)	0.28
REV6	1983-2024	5.71	(5.11)	0.25	(14.63)	0.30
NEWS	2000-2023	2.37	(1.47)	0.26	(12.42)	0.35
EW_FUND	1965-2024	5.91	(9.19)	0.24	(23.07)	0.43
RES_MOM	1965-2024	8.91	(10.05)	0.39	(27.60)	0.51
PTH	1927-2024	-5.67	(-3.74)	0.94	(44.97)	0.63
HTP	1927-2024	8.68	(5.81)	0.13	(6.44)	0.03
EW_ANCHOR	1927-2024	1.50	(2.98)	0.54	(77.10)	0.84
IND_MOM	1927-2024	7.21	(6.66)	0.42	(27.76)	0.40
ETNIC	2000-2023	6.62	(2.18)	0.97	(24.91)	0.68
C_ANLY	1983-2024	14.65	(5.87)	0.19	(4.86)	0.04
EW_NETWORK	1927-2024	8.13	(8.04)	0.42	(30.38)	0.44
F_MOM	1927-2024	9.07	(4.26)	0.31	(10.66)	0.09
EW_ALL	1927-2024	4.83	(8.61)	0.49	(63.72)	0.78

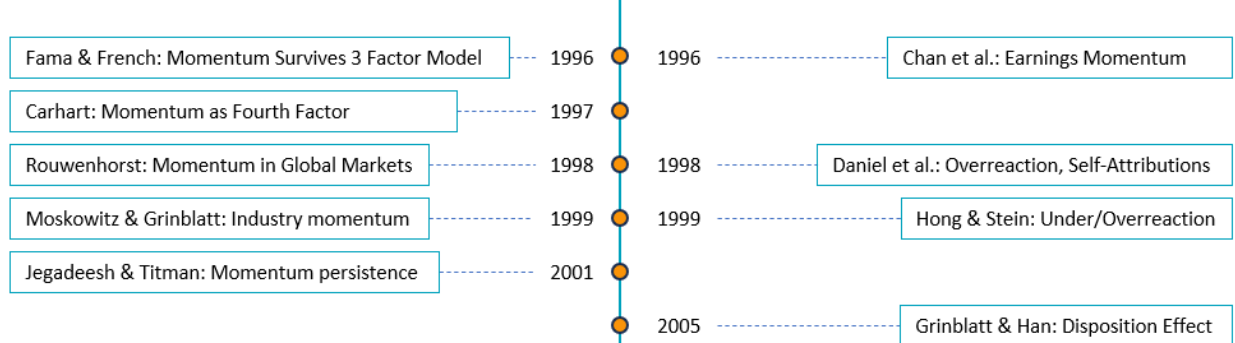
Table 5: **Momentum risk measures.** This table includes performance measures for the long-short momentum portfolios using US data. The results are included for the value-weighted (VW) and equal-weighted (EW) portfolios. Avg:= Annualized average returns (in %); Vol:= Annualized volatility (in %); Skew:= Skewness; Kurt:= Kurtosis; and MD:= Maximum drawdown (in %).

Char	Sample	VW					EW				
		Avg	Vol	Skew	Kurt	MD	Avg	Vol	Skew	Kurt	MD
PRICE_MOM	1927-2024	9.24	21.21	-2.36	22.08	-88.41	9.75	20.71	-3.98	38.40	-95.16
SUE	1965-2024	2.58	8.28	0.18	4.20	-34.29	8.61	6.09	-1.03	8.65	-13.21
REV6	1983-2024	5.46	11.09	0.17	5.35	-33.42	8.31	8.52	-2.15	16.28	-42.41
NEWS	2000-2023	1.15	8.72	-0.80	4.46	-36.57	4.22	9.72	-1.44	6.53	-27.80
EW_FUND	1965-2024	3.88	7.96	0.22	5.02	-22.20	8.63	6.46	-1.54	9.95	-22.40
RES_MOM	1965-2024	7.56	10.65	-0.25	4.13	-44.97	13.38	9.68	-1.59	15.04	-31.84
PTH	1927-2024	0.17	19.94	-3.46	33.39	-98.72	3.52	24.54	-3.87	31.87	-99.75
HTP	1927-2024	9.05	16.75	0.50	7.67	-65.09	9.97	14.88	2.01	21.37	-53.53
EW_ANCHOR	1927-2024	4.61	11.38	-1.15	14.52	-57.69	6.75	12.19	-3.34	34.40	-70.85
IND_MOM	1927-2024	6.73	14.73	-0.42	9.92	-61.99	11.26	13.67	-0.20	11.72	-54.22
ETNIC	2000-2023	9.79	24.86	0.01	5.60	-48.16	13.59	26.33	-1.30	15.36	-58.33
C_ONLY	1983-2024	7.55	18.49	0.21	4.60	-47.91	16.58	16.32	0.45	7.92	-36.37
EW_NETWORK	1927-2024	8.13	14.30	-0.45	12.35	-61.99	12.26	13.24	-0.81	17.66	-45.19
F_MOM	1927-2024	6.45	18.13	0.08	15.32	-71.89	12.12	21.82	2.68	46.09	-80.32
EW_ALL	1927-2024	6.35	11.57	-1.82	20.02	-56.84	9.65	11.62	-3.38	38.16	-51.96
RM_MOM	1927-2024	17.87	20.48	-0.14	0.97	-49.37	25.35	23.05	-0.54	2.03	-60.02
RM_MOM_SL	1927-2024	9.36	15.33	-0.35	5.28	-54.64	11.25	16.55	-1.05	15.62	-66.03

1970s – 1990s: Momentum Discovery



1995 - 2005: Momentum Robustness & Drivers



Post 2005: Beyond Price Momentum

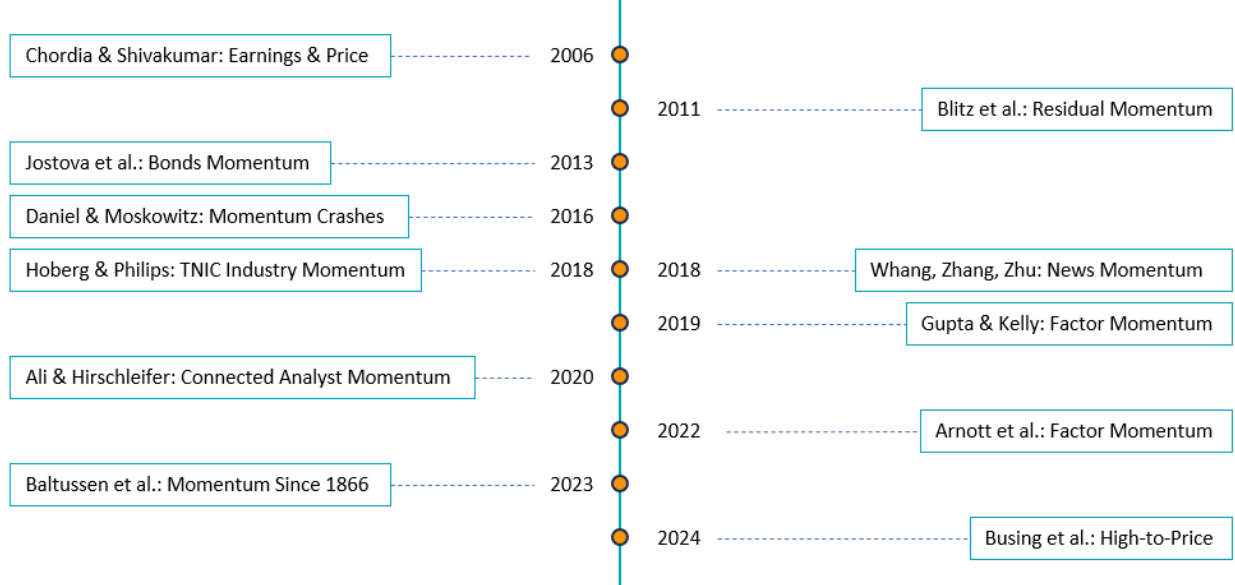


Figure 1: Timeline of Momentum Research.

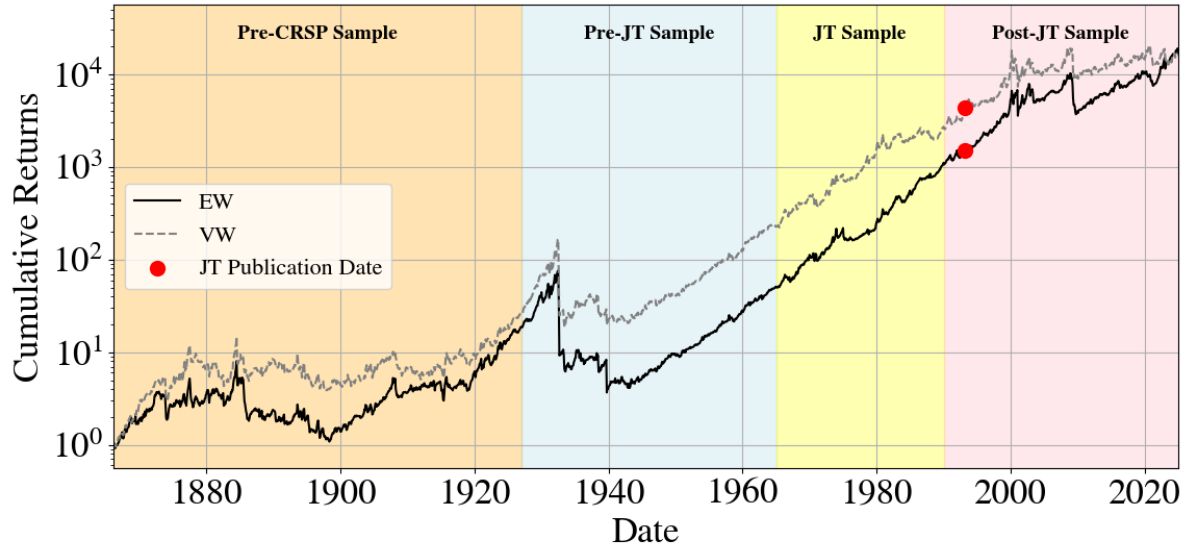
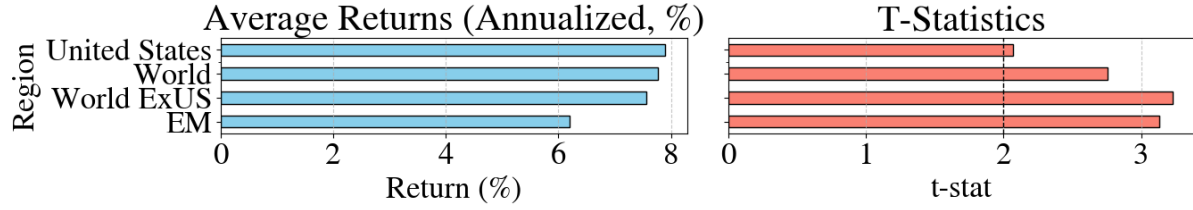
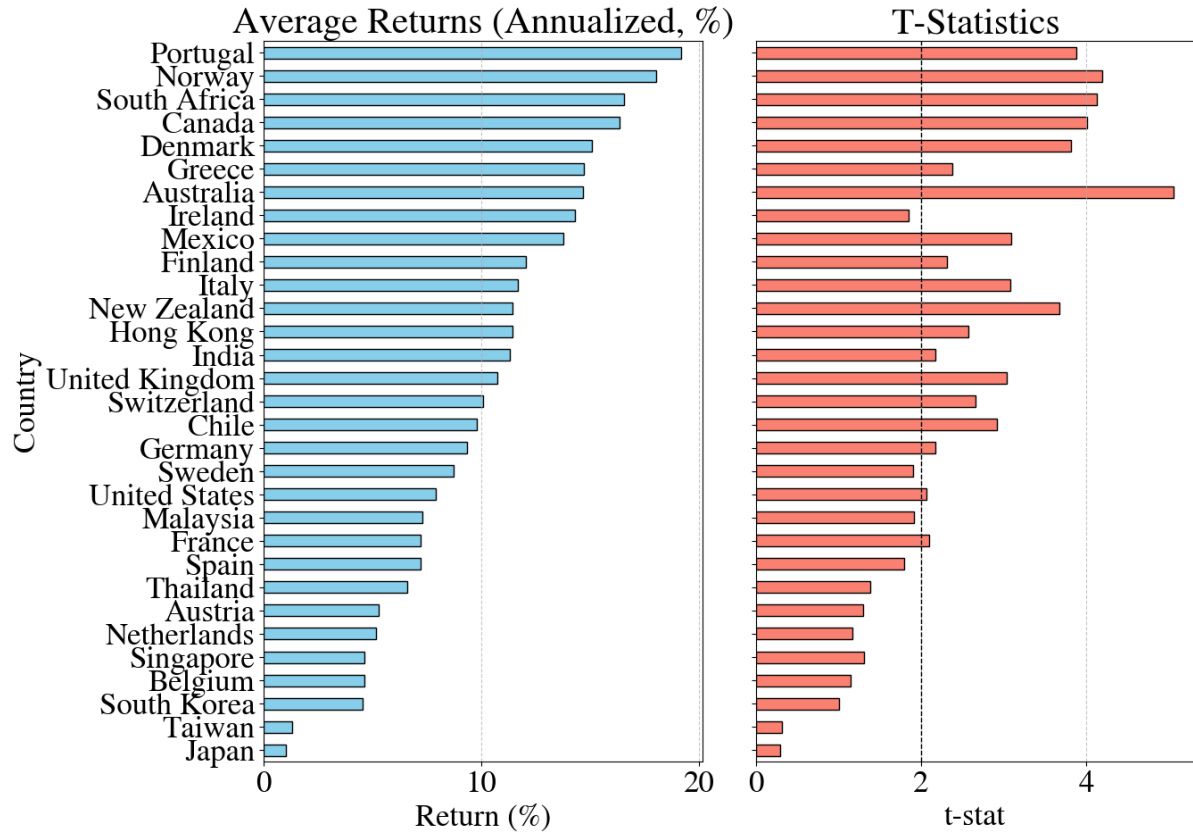


Figure 2: **Price momentum evolution in the US.** This figure shows the monthly cumulative returns of a price momentum strategy on US stocks from January 1866 to December 2024. The strategy entails a quintile long-short price momentum strategy. The plot displays equal-weighted (EW) and value-weighted (VW) portfolio returns on a log scale. It also highlights the publication month of (Jegadeesh & Titman, 1993) (March 1993). We distinguish four subsamples: Pre-CRSP (1866-1926), Pre-JT (1927-1964), JT (1965-1989), and Post-JT (1990-2024). The JT Sample denotes the sample period that was used by (Jegadeesh & Titman, 1993).



(a) Regional price Momentum.



(b) Country-level price Momentum.

Figure 3: **Regional and country-level price momentum results.** This figure shows the average annualized returns and corresponding t-statistics of the long-short quintile price momentum portfolios in 31 different countries from 1990 to 2024. 15 countries were excluded from this plot due to insufficient observations at the start of this sample.

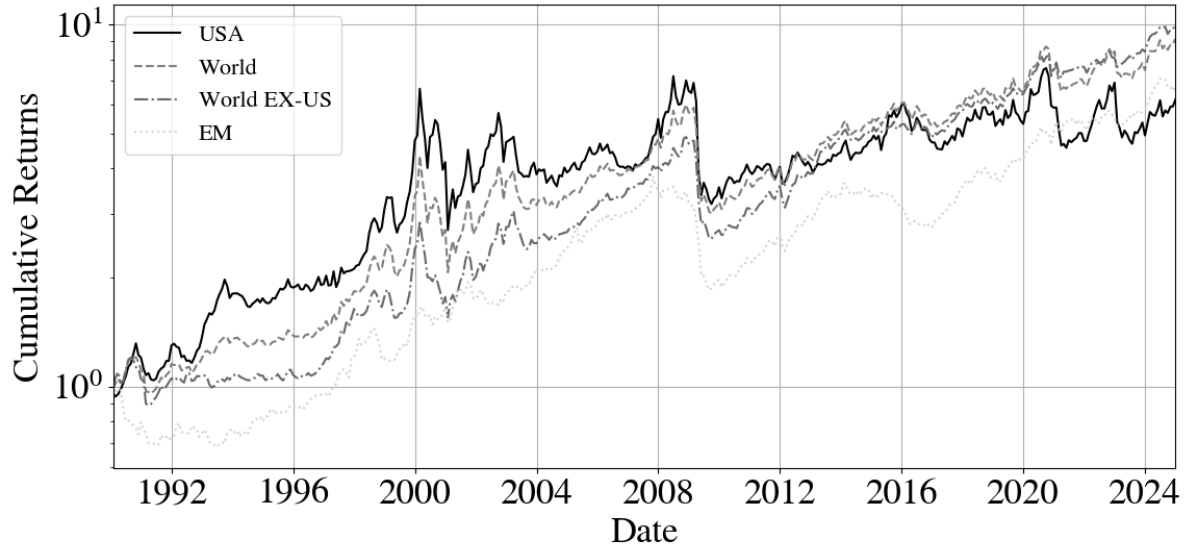


Figure 4: **Regional price momentum.** This figure shows the monthly cumulative returns of price momentum from 1990 to 2024 in four regions: The USA, World, World Ex-US, and Emerging Markets (EM). The returns are displayed on a log scale. We first computed the price Momentum quintile long-short portfolio returns for a total of 46 countries (value-weighted). The World portfolio returns comprise the weighted portfolio returns over all 46 countries, 23 of which are EM countries. The country returns are weighted using the 1-month lagged estimated market cap of each country as provided by (Jensen et al., 2023).

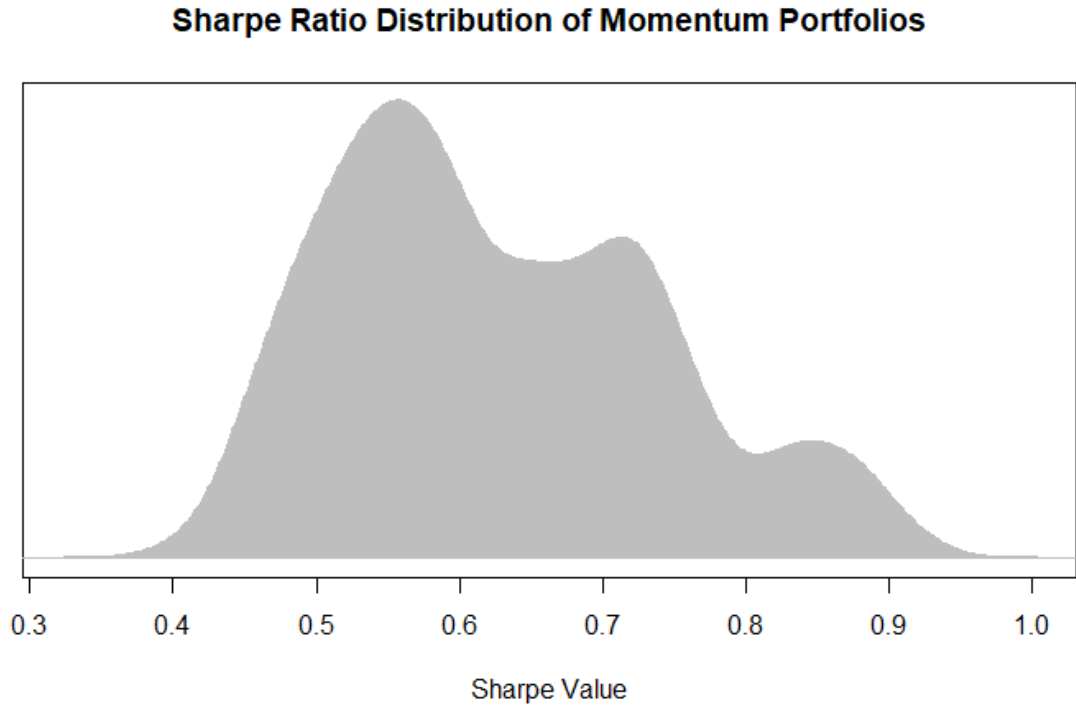


Figure 5: **Sharpe Ratio Distribution of Momentum Portfolios** This figure shows the distribution of Sharpe ratios across 4,096 factor construction specifications. We show Sharpe ratios for the long-short portfolio. The sample period is January 1972 to December 2023. .

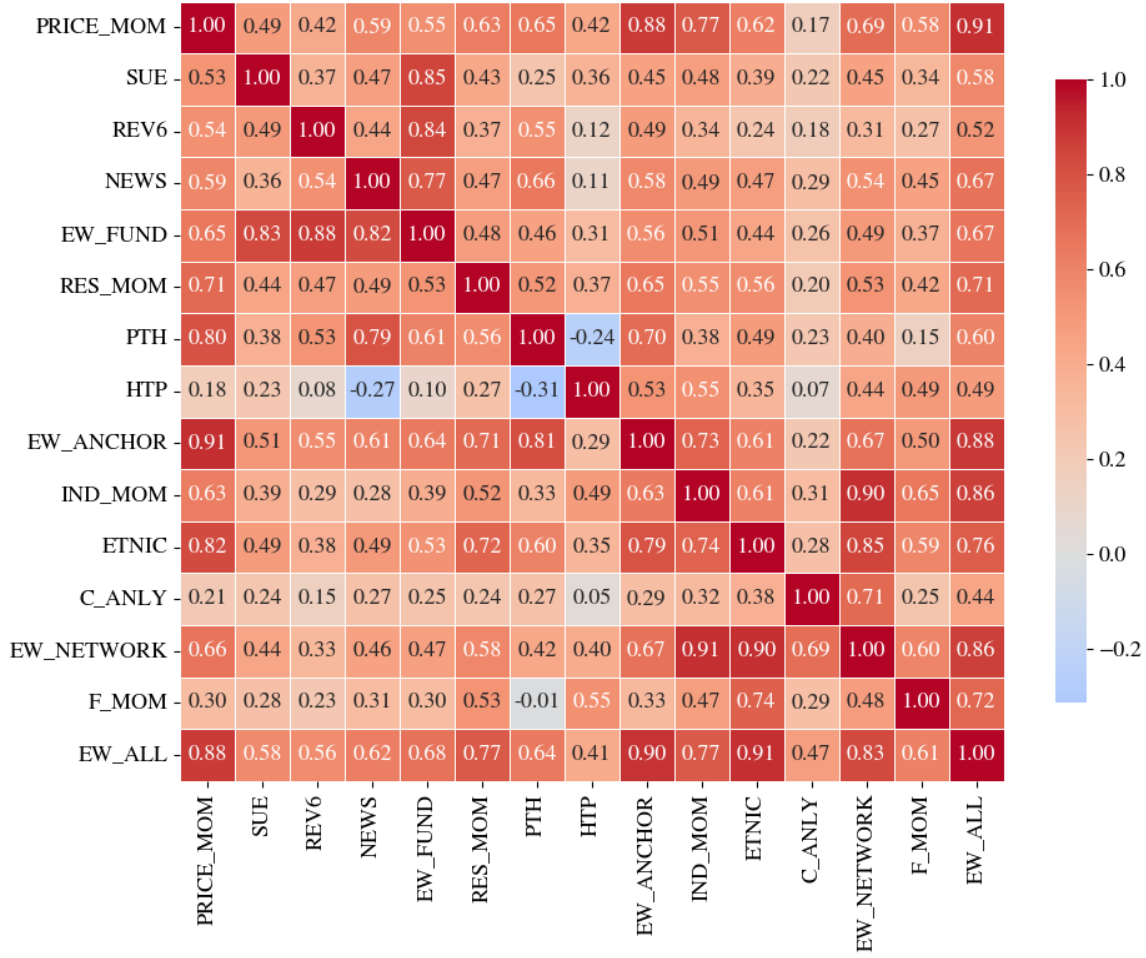


Figure 6: **Correlations of momentum strategies.** The upper right and lower left triangles in this figure show the pairwise Pearson correlations of the value-weighted and equal-weighted strategies, respectively. The pairwise correlations are computed over the maximum available sample period for each pair of characteristics.

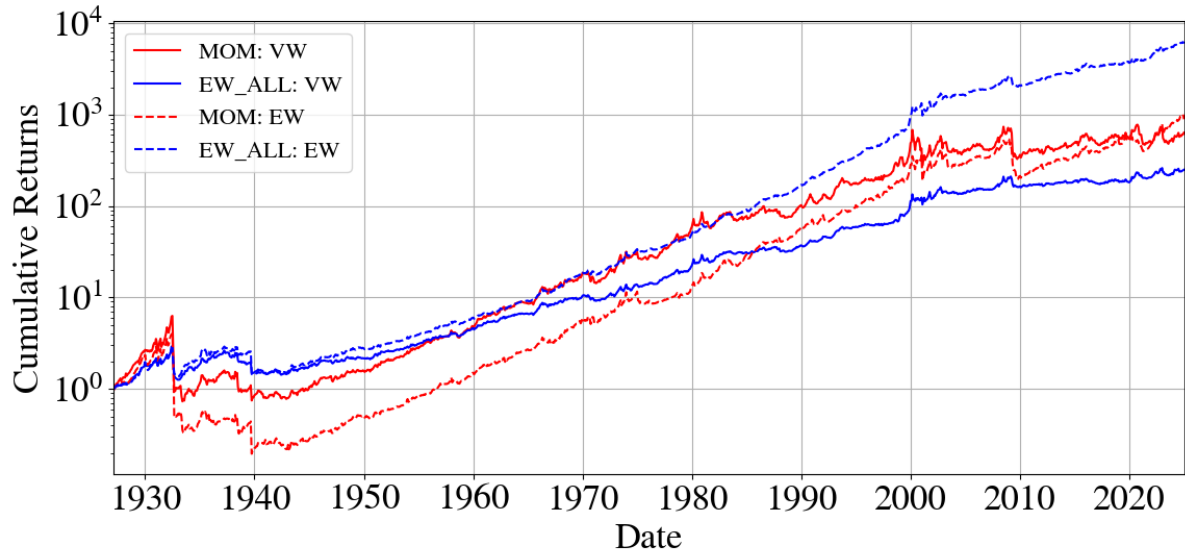


Figure 7: **Equal-weighted combination of momentum strategies.** The cumulative returns are shown on a log-scale over the full sample (1927–2024). The shown EW_ALL strategies are the average returns of price momentum and the ten alternative momentum strategies shown in Table 3.